Name: Debarshi Das Adhikary
Roll no: 255hsbb011
Variant call pipeline

OVERALL WORKFLOW
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Phase 1: Quality Control & Alignment

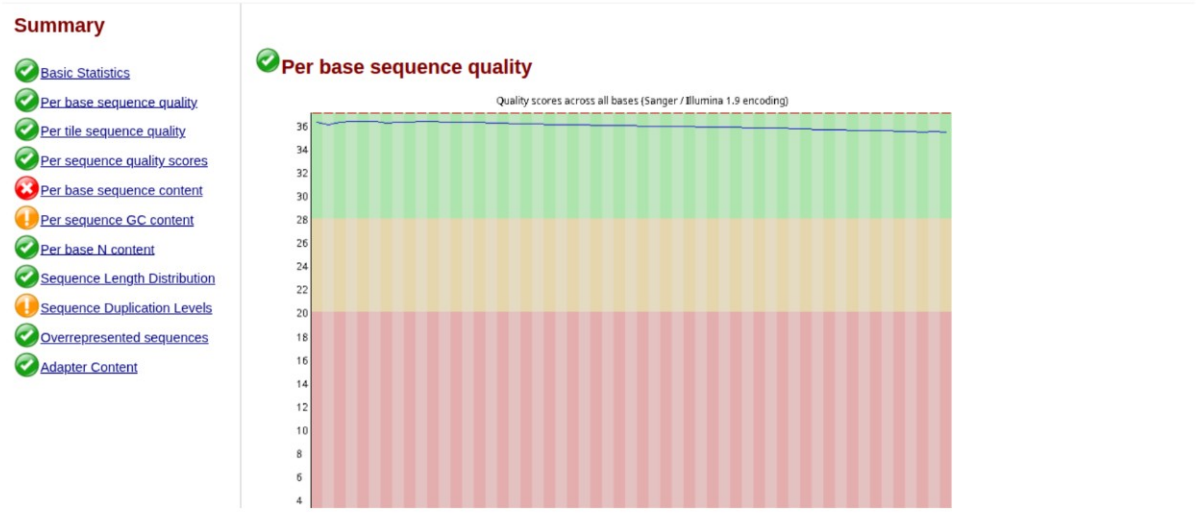
1. Quality Control

Run FastQC to inspect base qualities, adapter contamination, and duplication levels:

Bash

```
fastqc conditionA_Rep2.fastq
```

fastqc conditionA_Rep2.fastq



2. Index the Reference Genome for BWA

Generate the BWA FM-index for the reference FASTA:

Bash

```
bwa index GCF_000146045.2_R64_genomic.fa
```

3. Align Reads with BWA-MEM (Include Read Groups)

GATK requires read groups (@RG) containing sample and library metadata (ID, SM, PL, LB):

Bash

```
bwa mem -R '@RG\tID:Flowcell.Lane\tSM:conditionA_Rep2\tPL:ILLUMINA\tLB:LibraryName' \
    GCF_000146045.2_R64_genomic.fa conditionA_Rep2.fastq > conditionA_Rep2.sam
```

4. Convert SAM to BAM

Compress the SAM alignment to BAM format:

Bash

```
samtools view -bS conditionA_Rep2.sam > conditionA_Rep2.bam
```

5. Coordinate-Sort the BAM File

Sort alignments by genomic coordinate:

Bash

```
samtools sort conditionA_Rep2.bam conditionA_Rep2.sorted
```

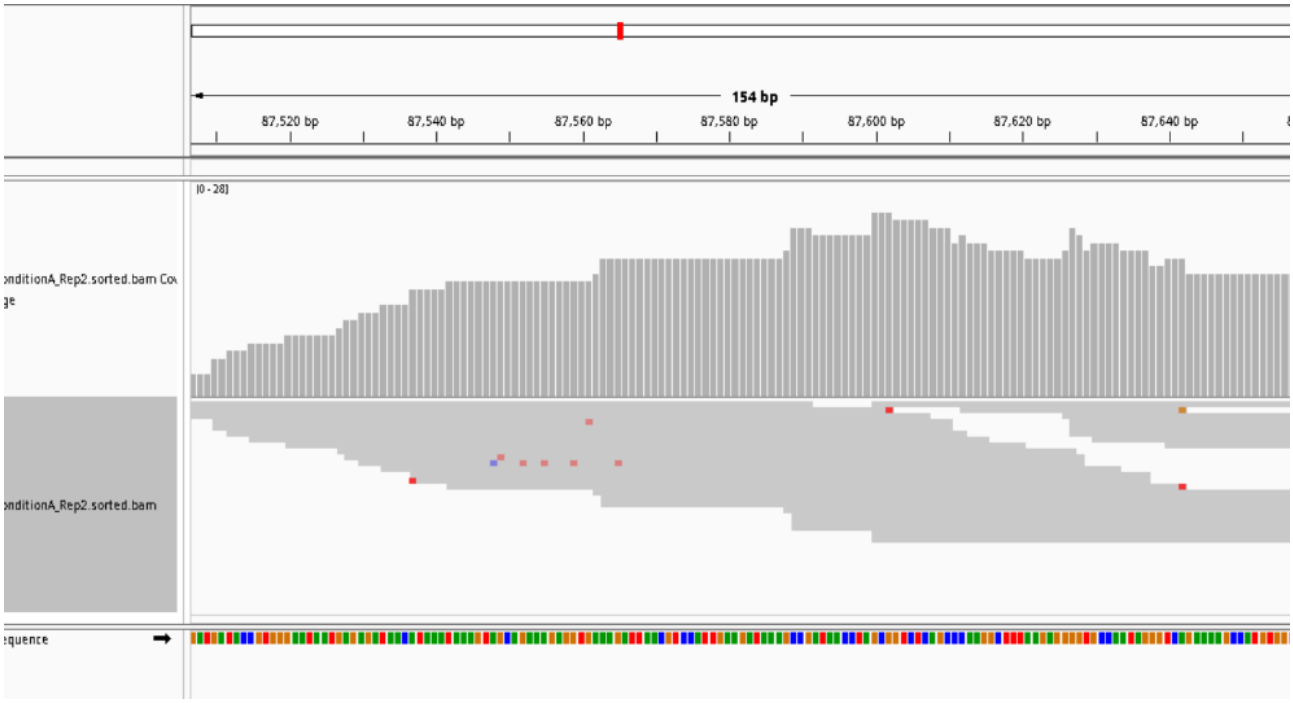
6. Index the Sorted BAM File

Generate the .bai index required for random access and visualization:

Bash

```
samtools index conditionA_Rep2.sorted.bam
```

IGV Visualization of Aligned Reads Before Variant Calling



The IGV view shows good read coverage across the selected genomic region, with reads aligned continuously to the reference and a few mismatching bases visible. These mismatches are potential variant sites that are subsequently evaluated by HaplotypeCaller and reported in the VCF.

Phase 2: Reference & Alignment Preparation for GATK

7. Build Reference Sequence Dictionary & FASTA Index

GATK requires a Picard sequence dictionary (.dict) and a FASTA index (.fai) alongside your reference FASTA:

Bash

```
# 7A: Create sequence dictionary
gatk CreateSequenceDictionary -R GCF_000146045.2_R64_genomic.fa

# 7B: Create samtools FASTA index
samtools faidx GCF_000146045.2_R64_genomic.fa
```

faidx - Creates a FASTA index.
This creates:
GCF_000146045.2_R64_genomic.fa.fai
This is another type of index, but this time it indexes the FASTA reference. So your reference will have:
GCF_000146045.2_R64_genomic.fa
GCF_000146045.2_R64_genomic.fa.fai
GCF_000146045.2_R64_genomic.dict

8. Mark PCR Duplicates

Mark duplicate reads to prevent technical PCR amplification artifacts from skewing variant allele depths:

Bash

```
picard MarkDuplicates \
    I=conditionA_Rep2.sorted.bam \
    O=conditionA_Rep2_markdup.bam \
    M=marked_dup_metrics.txt

# Index the markdup BAM
samtools index conditionA_Rep2_markdup.bam
```

Phase 3: Variant Calling & Quality Recalibration (BQSR)

9. Initial Variant Calling (Draft Set for BQSR)

HaplotypeCaller examines the reads in the BAM file, performs local assembly around regions that may contain variation, and calculates the likelihood of different haplotypes. It then reports likely SNPs and small insertions/deletions.

Bash

```
gatk HaplotypeCaller \
    -R GCF_000146045.2_R64_genomic.fa \
    -I conditionA_Rep2_markdup.bam \
    -O conA_variants.vcf
```

10. Base Quality Score Recalibration (BQSR)

Model systematic errors across cycle, read group, and context:

Bash

```
gatk BaseRecalibrator \
    -R GCF_000146045.2_R64_genomic.fa \
    -I conditionA_Rep2_markdup.bam \
    --known-sites conA_variants.vcf \
    -O BQSR_out_conA.table
```

Output : BQSR_out_conA.table- It is a recalibration table containing information about systematic biases in the reported base-quality scores.

11. Apply BQSR

Rewrite the BAM file with adjusted, calibrated base qualities:

Bash

```
gatk ApplyBQSR \
    -R GCF_000146045.2_R64_genomic.fa \
    -I conditionA_Rep2_markdup.bam \
    --bqsr-recal-file BQSR_out_conA.table \
    -O conA_BQSR_applied_reads.bam
```

The reads themselves are not being realigned.
The base quality scores associated with the bases are recalibrated according to the information in the

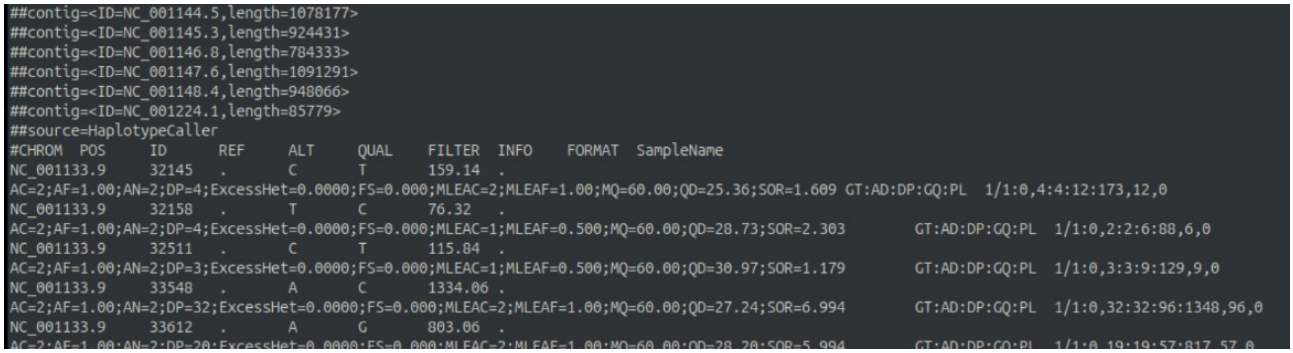
BQSR table.

12. Final Variant Calling

Call definitive variants on the recalibrated BAM:

Bash

```
gatk HaplotypeCaller \
    -R GCF_000146045.2_R64_genomic.fa \
    -I conA_BQSR_applied_reads.bam \
    -O conA_variants_final.vcf
```



```
##contig=<ID=NC_001144.5,length=1078177>
##contig=<ID=NC_001145.3,length=924431>
##contig=<ID=NC_001146.8,length=784333>
##contig=<ID=NC_001147.6,length=1091291>
##contig=<ID=NC_001148.4,length=948066>
##contig=<ID=NC_001224.1,length=85779>
##source=HaplotypeCaller
#CHROM  POS     ID      REF     ALT     QUAL    FILTER  INFO    FORMAT  SampleName
NC_001133.9     32145   .       C       T       159.14  .
AC=2;AF=1.00;AN=2;DP=4;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=60.00;QD=25.36;SOR=1.609 GT:AD:DP:GQ:PL  1/1:0,4:4:12:173,12,0
NC_001133.9     32158   .       T       C       76.32   .
AC=2;AF=1.00;AN=2;DP=4;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=28.73;SOR=2.303       GT:AD:DP:GQ:PL  1/1:0,2:2:6:88,6,0
NC_001133.9     32511   .       C       T       115.84  .
AC=2;AF=1.00;AN=2;DP=3;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=30.97;SOR=1.179       GT:AD:DP:GQ:PL  1/1:0,3:3:9:129,9,0
NC_001133.9     33548   .       A       C       1334.06 .
AC=2;AF=1.00;AN=2;DP=32;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=60.00;QD=27.24;SOR=6.994       GT:AD:DP:GQ:PL  1/1:0,32:32:96:1348,96,0
NC_001133.9     33612   .       A       G       803.06  .
AC=2;AF=1.00;AN=2;DP=20;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=60.00;QD=28.20;SOR=5.994       GT:AD:DP:GQ:PL  1/1:0,19:19:57:817,57,0
```

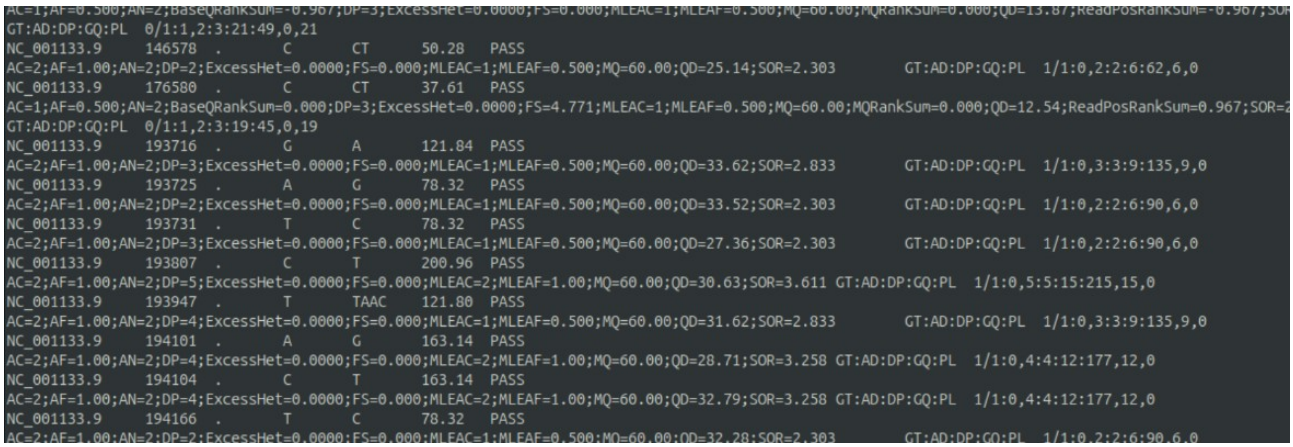
Phase 4: Variant Filtering & Annotation

13. Hard-Filter Low-Quality Variants

Flag low-quality SNPs and indels using standard GATK hard filters:

Bash

```
gatk VariantFiltration \
    -R GCF_000146045.2_R64_genomic.fa \
    -V conA_variants_final.vcf \
    --filter-expression "QD < 10.0" --filter-name "LowQD" \
    --filter-expression "FS > 60.0" --filter-name "HighFS" \
    --filter-expression "MQ < 40.0" --filter-name "LowMQ" \
    --filter-expression "MQRankSum < -12.5" --filter-name "LowMQRankSum" \
    -O conA_variants_final.filtered.vcf
```



```
AC=1;AF=0.500;AN=2;BaseQRankSum=-0.967;DP=3;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;MQRankSum=0.000;QD=13.87;ReadPosRankSum=-0.967;SOR
GT:AD:DP:GQ:PL  0/1:1,2:3:21:49,0,21
NC_001133.9     146578  .       C       CT      50.28   PASS
AC=2;AF=1.00;AN=2;DP=2;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=25.14;SOR=2.303       GT:AD:DP:GQ:PL  1/1:0,2:2:6:62,6,0
NC_001133.9     176580  .       C       CT      37.61   PASS
AC=1;AF=0.500;AN=2;BaseQRankSum=0.000;DP=3;ExcessHet=0.0000;FS=4.771;MLEAC=1;MLEAF=0.500;MQ=60.00;MQRankSum=0.000;QD=12.54;ReadPosRankSum=0.967;SOR=2
GT:AD:DP:GQ:PL  0/1:1,2:3:19:45,0,19
NC_001133.9     193716  .       G       A       121.84  PASS
AC=2;AF=1.00;AN=2;DP=3;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=33.62;SOR=2.833       GT:AD:DP:GQ:PL  1/1:0,3:3:9:135,9,0
NC_001133.9     193725  .       A       G       78.32   PASS
AC=2;AF=1.00;AN=2;DP=2;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=33.52;SOR=2.303       GT:AD:DP:GQ:PL  1/1:0,2:2:6:90,6,0
NC_001133.9     193731  .       T       C       78.32   PASS
AC=2;AF=1.00;AN=2;DP=3;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=27.36;SOR=2.303       GT:AD:DP:GQ:PL  1/1:0,2:2:6:90,6,0
NC_001133.9     193807  .       C       T       200.96  PASS
AC=2;AF=1.00;AN=2;DP=5;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=60.00;QD=30.63;SOR=3.611 GT:AD:DP:GQ:PL  1/1:0,5:5:15:215,15,0
NC_001133.9     193947  .       T       TAAC    121.80  PASS
AC=2;AF=1.00;AN=2;DP=4;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=31.62;SOR=2.833       GT:AD:DP:GQ:PL  1/1:0,3:3:9:135,9,0
NC_001133.9     194101  .       A       G       163.14  PASS
AC=2;AF=1.00;AN=2;DP=4;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=60.00;QD=28.71;SOR=3.258 GT:AD:DP:GQ:PL  1/1:0,4:4:12:177,12,0
NC_001133.9     194104  .       C       T       163.14  PASS
AC=2;AF=1.00;AN=2;DP=4;ExcessHet=0.0000;FS=0.000;MLEAC=2;MLEAF=1.00;MQ=60.00;QD=32.79;SOR=3.258 GT:AD:DP:GQ:PL  1/1:0,4:4:12:177,12,0
NC_001133.9     194166  .       T       C       78.32   PASS
AC=2;AF=1.00;AN=2;DP=2;ExcessHet=0.0000;FS=0.000;MLEAC=1;MLEAF=0.500;MQ=60.00;QD=32.28;SOR=2.303       GT:AD:DP:GQ:PL  1/1:0,2:2:6:90,6,0
```

14. Check Database & Harmonize Chromosome Names

SnpEff requires chromosome identifiers to match its database entries exactly.

1. **Verify SnpEff Database Names**:

   Bash

   ```
   snpEff dump R64-1-1.99 | head -n 50
   ```

2. **Check Your VCF Chromosome Names**:

   Bash

   ```
   grep -v "^#" conA_variants_final.filtered.vcf | cut -f1 | sort -u
   ```

3. **Map and Rename Using bcftools** (Create a tab-delimited mapping file chr_map.txt if NCBI accessions like NC_001133.9 need conversion to SnpEff names like I, chrI, etc.):

   Bash

   ```
   bcftools annotate --rename-chrs chr_map.txt \
     -o conA_variants_snpEff_input.vcf -Ov \
     conA_variants_final.filtered.vcf
   ```

15. Functional Annotation with SnpEff

Annotate variant impact and generate HTML quality/effect statistics:

Bash

```
snpEff -stats conA_snpEff_summary.html \
  R64-1-1.99 \
  conA_variants_snpEff_input.vcf > conA_variants_annotated.vcf
```

SnpEff: Variant analysis

Contents

Summary
Variant rate by chromosome
Variants by type
Number of variants by impact
Number of variants by functional class
Number of variants by annotation
Quality histogram
InDel length histogram
Base variant table
Transition vs transversions (ts/tv)
Allele frequency
Allele Count
Codon change table
Amino acid change table
Chromosome variants plots
Details by gene

Summary

Genome	R64-1-1.99
Date	2026-09-16 10:52
SnpEff version	SnpEff 5.3a (build 2025-09-02 10:24), by Pablo Cingolani
Command line arguments	SnpEff R64-1-1.99 conA_renamed.vcf
Warnings	1,969
Errors	0
Number of lines (input file)	8,283
Number of variants (before filter)	8,290
Number of non-variants (i.e. reference equals alternative)	0
Number of variants processed (i.e. after filter and non-variants)	8,287
Number of known variants (i.e. non-empty ID)	0 (0%)
Number of multi-allelic VCF entries (i.e. more than two alleles)	7
Number of annotations	57,055

Missense / Silent ratio: 0.4553

Number of annotaitons and region counts

Annotation

Type (alphabetical order)	Count	Percent
conservative_inframe_deletion	5	0.009%
conservative_inframe_insertion	8	0.014%
disruptive_inframe_deletion	25	0.044%
disruptive_inframe_insertion	10	0.018%
downstream_gene_variant	25,607	44.873%
frameshift_variant	96	0.168%
intergenic_region	1,324	2.32%
intragenic_variant	1	0.002%
intron_variant	10	0.018%
missense_variant	2,160	3.785%
non_coding_transcript_exon_variant	27	0.047%
splice_acceptor_variant	1	0.002%
splice_donor_variant	1	0.002%
splice_region_variant	6	0.011%
start_lost	3	0.005%
stop_gained	9	0.016%
stop_lost	1	0.002%
stop_retained_variant	5	0.009%
synonymous_variant	4,743	8.312%
upstream_gene_variant	23,023	40.345%

Region

Type (alphabetical order)	Count	Percent
DOWNSTREAM	25,607	44.881%
EXON	7,085	12.418%
INTERGENIC	1,324	2.321%
INTRON	8	0.014%
SPLICE_SITE_ACCEPTOR	1	0.002%
SPLICE_SITE_DONOR	1	0.002%
SPLICE_SITE_REGION	5	0.009%
TRANSCRIPT	1	0.002%
UPSTREAM	23,023	40.352%

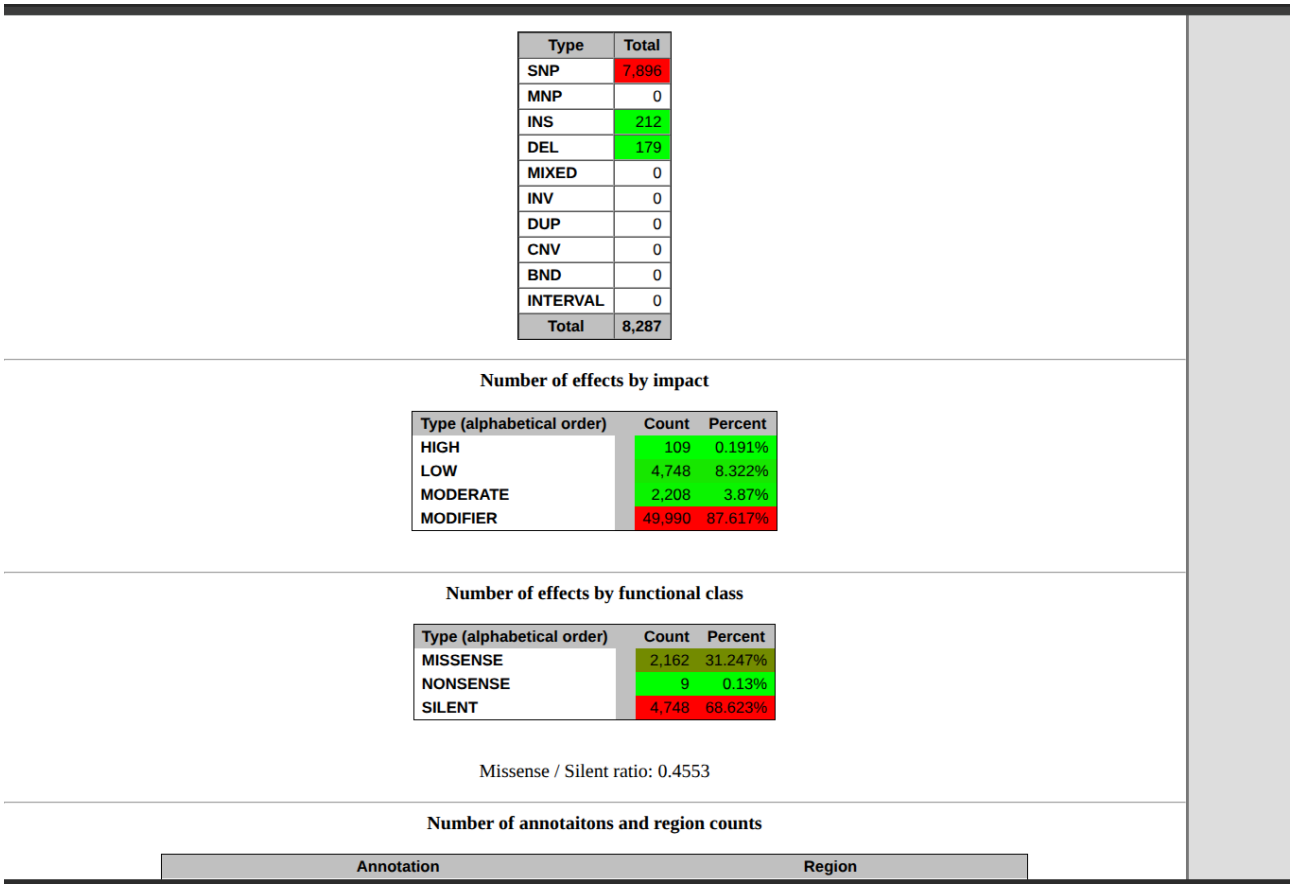


Quality:

Min	30
Max	3,481
Mean	242.939

SnpEff Summary Interpretation

- **Variant Count & Types:** We identified **8,805 variants** in total, dominated by SNPs (8,309), along with 273 insertions and 223 deletions. Across the 12.16 Mb yeast genome, that averages about **1 variant every ~1.38 kb**.
- **Genomic Distribution:** Chromosome XIII had the most variants (1,278), while the mitochondrial genome had the fewest (only 5). Chromosome V showed the highest variant density.
- **Functional Impact:** The vast majority of variants fall into non-coding or mild categories—**87.6% are MODIFIER** and **8.1% are LOW** impact. Only **175 variants (~0.3%)** are predicted to have a **HIGH impact** (likely disrupting gene function).
- **Coding Effects:** Among coding changes, about **two-thirds (67.6%) are silent/synonymous**, meaning they don't alter the protein sequence. There are **2,344 missense** mutations and just **15 nonsense** (premature stop) mutations.
- **Key Takeaway:** Most detected variants are benign or non-disruptive. For downstream biological analysis, prioritize the **175 HIGH-impact** and **2,344 missense** variants, while double-checking the ~2,100 annotation warnings.



Type	Total
SNP	7,896
MNP	0
INS	212
DEL	179
MIXED	0
INV	0
DUP	0
CNV	0
BND	0
INTERVAL	0
Total	8,287

Number of effects by impact

Type (alphabetical order)	Count	Percent
HIGH	109	0.191%
LOW	4,748	8.322%
MODERATE	2,208	3.87%
MODIFIER	49,990	87.617%

Number of effects by functional class

Type (alphabetical order)	Count	Percent
MISSENSE	2,162	31.247%
NONSENSE	9	0.13%
SILENT	4,748	68.623%

Missense / Silent ratio: 0.4553

Number of annotaitons and region counts

Annotation	Region